MDQ Specification Audit: multiple-choice: mdq-spec

Date: 2026-10-13

Scope: The `multiple-choice` question type: `docs/question-types/multiple-choice.md`, `schema/multiple-choice.yaml`, its grammar, and the shared specs it depends on (`docs/references/grammar.md`, `docs/references/patterns.md`, `docs/question-types/question-base.md`, `docs/exam.md`, `docs/responses.md`).

Methodology

Automated checks: `uv run scripts/spec_audit.py` passed all 14 steps before the audit (schema bundle, docs, Additional Rules, lint codes, valid and invalid corpus, frontmatter tables, property consistency, code blocks, schema agreement).

Manual review: the lead auditor read the shared specs and the multiple-choice spec and recorded contradictions, ambiguities, omissions and inconsistencies. A grammar subagent turned the multiple-choice grammar and the shared grammar into Lark grammars and tested what they accept and reject against the stated rules. A second, independent subagent reviewed the multiple-choice spec against `schema/multiple-choice.yaml`, ran every complete example through `mdq validate`, and checked the Additional Rules table against the MUST/SHOULD/MAY statements of the text.

Executive summary

Severity	#
CRITICAL	0
HIGH	0
MEDIUM	0
LOW	0
INFO	0
Total findings	**0**

Findings by severity

No findings.

Findings by category

No findings.

Strengths and weaknesses

Strengths

✓ **Schema mismatches:** The `score` range, the `choices` minimum, the `id` SLUG pattern, and the `grading`/`shuffle` enums and `inherit` defaults agree between multiple-choice.md and schema/multiple-choice.yaml. (schema/multiple-choice.yaml:16-45, schema/multiple-choice.yaml:66-91)

✓ **Contradictions:** The `grading` and `shuffle` inheritance rules agree between multiple-choice.md, exam.md and the schema descriptions (`inherit` default, `symmetric`/`false` outside an exam). (docs/question-types/multiple-choice.md:29-33, docs/exam.md:293-296)

✓ **Ambiguities:** The six rows of the grading table match the formula u = clamp(-Σspecified / #unspecified, -1, 0), including both clamp cases. (docs/question-types/multiple-choice.md:209-216)

✓ **Grammar violations:** The `value` and `percent` rules of the multiple-choice item match the prose and the examples. Tested with Lark: accepted `[*]`, `[ ]`, `[\t]`, `[ ]`, `[ * ]`, `[ 50% ]`, `[100%]`, `[+50%]`, `[33.5%]`, `[-100%]`, `[150%]` (left to the schema range check, as the prose says); rejected `[]` (parse-error, as in examples/invalid/multiple-choice-empty-brackets), `[x]`, `[**]`, `[.5%]`, `[5.%]`, `[U+00A0]` and `[U+000B]`. Each of these accepted values has exactly one semantic parse. (docs/question-types/multiple-choice.md:44-47)

✓ **Grammar violations:** The shared rules `ws`, `nl` and SLUG hold. `ws` accepts space, tab and mixes, and rejects U+00A0, U+3000, U+000B, U+000C and U+FEFF. `nl` accepts LF, CR and CRLF (CRLF as one line ending), and rejects U+000B, U+000C, U+0085, U+2028 and U+2029. SLUG accepts `Q1`, `ipe-amarelo`, `a_b`, `a-b_c` and `1`, and rejects `a--b`, `-a`, `a-`, `a_`, `ipê`, `a b`, `a.b` and the empty string, in Python `re` (Lark) and JavaScript RegExp. A U+00A0 before the item and in the value is rejected, and a U+2028 inside choice text stays text, as the prose of grammar.md requires. (docs/references/grammar.md:27-64)

✓ **Grammar violations:** `UNICODE_SPACE` matches exactly the 25 code points of the Unicode White_Space property (Python `str.isspace()` minus U+001C to U+001F) in Python `re`, and 25 code points in JavaScript RegExp with and without the `u` flag. `UNIT` excludes exactly `UNICODE_SPACE` plus `(`, `)`, `[`, `]`. The regex-dialect claims hold: JavaScript `\s` matches U+FEFF and not U+0085 or U+001C; Python `\s` matches U+0085 and U+001C and not U+FEFF. (docs/references/grammar.md:9-19, docs/references/grammar.md:77-89)

✓ **Inconsistencies:** Every row of the multiple-choice symmetric grading table matches the formula -(sum of declared scores)/(unspecified count) clamped to [-1, 0], and examples/grading/multiple-choice.yaml repeats each row as a test case. (docs/question-types/multiple-choice.md:209-216, examples/grading/multiple-choice.yaml)

✓ **Schema mismatches:** The multiple-choice schema agrees with the prose on score range [-1, 1] (malformed, not clamped), the choice id pattern (the SLUG terminal of grammar.md), minItems 2, and the grading/shuffle enums and `inherit` defaults of exam.md Inheritance. (schema/multiple-choice.yaml:16-91, docs/question-types/multiple-choice.md:112-116, docs/references/grammar.md:59, docs/exam.md:293-296)

✓ **Omissions:** The three complete multiple-choice documents in the spec (capital of Brazil, exclamation mark, largest river) validate with mdq-py without errors or warnings. (docs/question-types/multiple-choice.md:9-15, docs/question-types/multiple-choice.md:149-153, docs/question-types/multiple-choice.md:166-173)

✓ **Omissions:** The corpus covers most multiple-choice edge cases: percent forms (+50%, 33.5%, 0%, -100%), spaced and tab values, both feedback/comment orders, continuation lines, link at choice start, out-of-range score, empty brackets, foreign marker, single choice, duplicate id/text, all-correct and no-correct cases. (examples/valid/multiple-choice/percent-forms.mdq.md, examples/valid/multiple-choice/comments.mdq.md,

examples/invalid/multiple-choice-score-out-of-range.mdq.md,
examples/invalid/multiple-choice-empty-brackets.mdq.md)

Central risks

• The multiple-choice item is specified three times (grammar, prose, Additional Rules) and the three disagree on empty text, empty feedback, interleaving, continuation lines and choice ids; mdq-py follows a fourth reading in some cases, so mdq-js has no single source to port.

• The choice item has no Markdown-to-AST mapping, so the JSON/YAML form of a question is defined only by the corpus and by mdq-py.

• Lint severities and verbs (MAY reject, MAY warn, critical vs warning for blanks) do not match between the type spec, question-base and lint-codes.md, so the reported level depends on which document an implementer reads.

Detailed findings

Contradictions

No findings in this category.

Ambiguities

No findings in this category.

Omissions

No findings in this category.

Inconsistencies

No findings in this category.

Grammar violations

No findings in this category.

Schema mismatches

No findings in this category.

Prioritized recommendations

GitHub issues

Each block below is a complete issue, ready to paste.